
Consensus statement on Brillouin light scattering microscopy of biological materials

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Table of Content:

1: Different BLS instruments & reporting recommendations.....	p2-p13
1.1. Spontaneous BLS.....	p2
<i>Single etalon spectrometers.....</i>	<i>p2</i>
<i>Multi-pass Fabry-Pérot interferometers.....</i>	<i>p2</i>
<i>VIPA-based spectrometers.....</i>	<i>p3</i>
<i>Crossed VIPAs.....</i>	<i>p3</i>
<i>Crossed VIPA-echelle grating.....</i>	<i>p3</i>
<i>Elastic suppression filters.....</i>	<i>p4</i>
<i>Gas absorption cell.....</i>	<i>p5</i>
<i>Etalon-based.....</i>	<i>p5</i>
<i>Radial VIPA.....</i>	<i>p6</i>
1.2. Stimulated BLS.....	p6
<i>Frequency domain SBS.....</i>	<i>p7</i>
<i>Impulsive SBS.....</i>	<i>p7</i>
<i>Time domain BLS.....</i>	<i>p7</i>
1.3. Heterodyne BLS.....	p10
1.4. Fiber probes.....	p11
2: Some useful equations for BLS microscopy.....	p16-p30
2.1. Fitting equations.....	p16
2.2. Frequency shift.....	p17
2.3. Linewidth.....	p19
2.4. Loss tangent.....	p21
2.5. Effect of Acoustic Heterogeneities.....	p21
2.6. Relation to other moduli.....	p22
2.7. Calculation of stiffness tensor.....	p24
2.8. Registration to accepted/standard values.....	p28
2.9. Determination of the refractive index and mass density.....	p30
Supplementary References.....	p31-p33

Supplementary Figures

Supplementary Figure 1: <i>Brillouin Light Scattering Spectrometer measurement schemes.....</i>	<i>p14</i>
Supplementary Figure 2: <i>Brillouin Light Scattering spectra of polystyrene beads.....</i>	<i>p15</i>
Supplementary Figure 3: <i>Stiffness Tensor.....</i>	<i>p24</i>
Supplementary Figure 4: <i>BLS Scattering Geometries.....</i>	<i>p26</i>
Supplementary Figure 5: <i>Stiffness Tensor Symmetries.....</i>	<i>p28</i>

1: Different BLS instruments & reporting recommendations

1.1. Spontaneous BLS

Spontaneous BLS, broadly refers to the case where the probed phonons are not being generated/stimulated by some additional means (e.g. by additional lasers or mechanical transducers). To this end there is typically only a single probing laser, from which the scattered light is analysed, to reveal the scattering from inherent thermal phonons. Currently spontaneous BLS techniques are exclusively confined to the frequency domain, namely they measure the frequency spectrum of the scattered light. In the following subsections, we describe some details and specifics to instruments used in spontaneous BLS measurements.

Single etalon spectrometers

The simplest setup for a Brillouin is a non-scanning single pass, single etalon system. This yields a standard Fabry-Perot set of fringes where the center is the zeroth order and each successive mode is a higher order with non-linear dispersion between orders. Orders are Rayleigh peaks; surrounding each order is the set of Brillouin Stokes and anti-Stokes pairs. In a single etalon system, the light has to be angularly dispersed through the etalon either by tilting or diverging the beam. A cylindrical lens can be used to compress the dispersing light to a line rather than illuminating the entire circular Fabry-Perot pattern. FSR range between orders requires calibration and linearization yet multiple orders can be used for spectroscopic information. Experimental parameters important to report with a single etalon system are the manufacturer, air-gapped (and the gas if different than air) or solid etalon, the flatness, free-spectral range, finesse, how many orders are used in analysis, and whether the spectra is linearized or not linearized.

Multi-pass Fabry-Pérot interferometer

A Tandem Fabry-Pérot spectrometer (TFP) relies on a tandem combination of two air spaced etalons whose spacing is precisely adjusted and controlled. By varying the mirror separation of the etalons, the device sequentially scans consecutive wavelengths. Multiple passes through the etalons are used to increase contrast and finesse. One popular example of this is the 6-pass TFP spectrometer TFP-2 HC, a commercial device developed by Table Stable Ltd., which can achieve a contrast of up to 10^{15} . This high contrast can suppress the elastic background in even strongly turbid samples¹.

The FSR can be varied between about 5 THz and 5 GHz, with a finesse of 100-120 giving a maximum resolution of about 50 MHz. The standard scan time is ~500 ms, but the speed and acquisition bin size can be modulated allowing for e.g. selected spectral windows to accumulate more photons quicker and thus for faster measurements. Using external timing sources, a commercial TFP-2 HC allows faster scanning up to a factor 3-5. A faster version of the TFP, capable of reaching full scan times of 30 ms without any loss of performance, has

been developed successfully by the company Table Stable Ltd. and will be commercialised soon.

A major advantage of the scanning Fabry-Perot is that it is an absolute instrument: the wavelength calibration is determined uniquely by the mirror spacing which is typically known to $\sim 0.1\%$ or better. As such no calibration standard is required, and under laboratory conditions such an instrument will stay stable for hours and even days. Transmission efficiency can be checked by comparison to known samples. When frequency locked to the laser wavelength, there is no theoretical limit on how long such an instrument can be held stable.

When publishing measurements, in addition to the spectrometer manufacturer, one should report the mirror spacing, pinhole sizes (entrance pinhole to spectrometer and exit pinhole to detector), scan amplitude(s), total recording time per channel, Quantum efficiency (QE) of the photon detector used, and a full description of the optics feeding the signal into the TFP.

VIPA-based spectrometers

VIPA (Virtual Imaged Phase Array) spectrometers are non-scanning systems that capture the entire BLS spectrum simultaneously. A VIPA is a type of Fabry-Perot etalon in which the upstream partial reflector is replaced by a high reflector, except for a small window through which the light is coupled. The dispersion of a VIPA is the same as that of a standard angularly dispersed etalon. While the latter has a throughput on the order of inverse finesse, the transmission of a VIPA approaches unity for a near diffraction-limited input along the dispersion axis (e.g. confocal collection or line scan geometry)². Consequently, depending on the excitation conditions and sample, VIPA spectrometers can acquire a Brillouin spectrum in a few tens of milliseconds. A FWHM resolution of a few 100 MHz and a sub-10 MHz precision are typical. The VIPA thickness can be determined within 0.1%, facilitating accurate absolute calibration. The simplest VIPA spectrometer setup consists of a collimated beam focused with a cylindrical lens into the entrance window of a VIPA, followed by an imaging lens and a camera to capture the spectrum. The contrast of a single VIPA being inherently limited to ~ 30 dB, several cross-dispersion configurations and elastic peak suppression strategies have been developed as described below.

Crossed VIPAs: To increase the contrast of spectrally-dispersive optical elements, cross-axis cascading of multiple elements³ offers an efficient option that has been used in many VIPA-based spectrometers. The principle of cross-axis cascading with VIPA etalons is straightforward: if a first stage of optical dispersion is aligned along a vertical direction, then the spectral pattern is also dispersed vertically. In this situation, any elastic scattering component beyond the contrast of the first spectral stage appears as a crosstalk signal along the vertical spectral axis. If a second stage of spectral dispersion is now aligned orthogonally to the first stage, the spectral pattern exiting the first stage enters the second etalon through the input window. Since both etalons disperse light in orthogonal directions, the overall spectral axis of the double-spectrometer lies along a diagonal direction, which is where the

Brillouin spectrum appears. Instead, the crosstalk elastic signal, which was spatially overlapped after the first stage, after the second stage, is mostly confined to the horizontal and vertical axis. This procedure has been shown to be repeatable for multiple stages⁴, but the double-stage configuration has proven to provide the best tradeoff between throughput and contrast. In most realizations, the elastic scattered light is physically blocked with an optical mask in an intermediate image plane or planes to avoid detector saturation. The crossed-VIPA architecture is compatible and often used with other improvements that have been demonstrated over the years such as apodization⁵, coronagraphy (Lyot stop)⁶ or diffraction masks⁷, leading to high-throughput contrast of ~80 dB. These combinations usually suffice for measuring e.g. isolated biological cells, but not for more opaque samples.

Crossed VIPA-echelle grating: As an alternative to crossed VIPAs, contrast enhancement through cross-dispersion can also be achieved with an echelle grating. The latter displaces the Brillouin signals away from the VIPA Lorentzian tails of the typically much stronger elastic signal. This results in a contrast on the order of 50 dB at the water BLS peak position and up to 60 dB beyond. The grating also separates otherwise overlapping VIPA orders, producing an unambiguous simultaneous range of several THz with minimal spectral artifacts. This enables the BLS and ultra-low frequency Raman measurement of a large range of samples in a snapshot. Since this configuration produces a 2D pattern on the camera, calibration involves mapping which pixels can be illuminated and determining to what wavelength they correspond based on the VIPA and grating dispersion equations. Commercial solutions are available employing such a crossed VIPA-echelle grating scheme, which when combined with an elastic suppression filter can achieve an effective contrast of 120 dB⁸. Alignment of the crossed VIPA-echelle grating scheme is similar to a single VIPA, since the grating is essentially alignment-free.

When publishing measurements on VIPA-based spectrometers additional parameters that should be reported are the VIPA FSR, the spectral interval measured on the detector array, the number of orders used to analyze the BLS peaks, and if relevant the model number in the case of a commercial instrument.

Elastic suppression filters: An alternative solution to cross-dispersion that has been used to gain visibility of the BLS spectral peaks, in primarily VIPA-based spectrometers, is through the employment of special high-rejection narrow-band optical filters to attenuate the elastic scattering light. In this case, the elastic background light is suppressed by a notch filter, relaxing the need for multiple optical passes and/or cascaded dispersive elements to achieve high spectral contrast at detection. Employing a sufficiently effective filter can allow one to employ only a single VIPA spectrometer to perform measurements on turbid media.

Several methods have been demonstrated to efficiently suppress the elastic background light with extinction ratio ranging from 30 to 50 dB. Commonly employed methods include the use of etalons^{9,10}, absorption cells¹¹ or interferometric schemes^{12,13}. Two popular filter types currently being employed are:

Gas absorption cell filters: Atomic and molecular line absorption filters can provide an efficient approach to suppress the elastic scattering in BLS microscopy. Since some of the most commonly used laser wavelengths for BLS microscopy (532 nm and 780 nm) are close to molecular absorption lines of molecular Iodine (532 nm) and atomic Rubidium (780 nm), the use of absorption lines from these to suppress elastic scattering can be used. Here the incident laser light can be tuned to match a selected absorption line and be actively stabilized and frequency locked to this line¹⁴. The advantages are the relatively low cost and easy maintenance of such a filter, which are independent of environmental conditions and beam divergence. The Doppler and collisional linewidth are insignificant for most practical conditions, and a high degree of attenuation can be achieved using multipass geometry or longer cell pathlengths^{15,16}.

Potential drawbacks of gas absorption cells include the presence of a number of absorption lines that may modify the transmission spectrum and thus final measured spectra (particularly the case in Iodine cells), residual emission from the cell, and a potentially large variation of transmission for a cell whose temperature is not actively stabilized. Some of these issues can be addressed through BLS measurements taken at multiple excitation wavelengths, at the expense of making the overall setup more complicated and less suitable for rapid acquisition¹⁷. Gas absorption cells also ideally require the use of laser lock-in scheme, which typically comes with an increased optical system complexity. When using gas absorption cells, it is recommended to report the chemical composition of the cell and its dimensions, the number of passes through the cell, the manufacturer and model of the cell, and the temperature of the cell when in use.

Etalon-based filters: A low-insertion loss, ultra-narrow notch filter can be achieved with a Fabry-Perot etalon tuned (angle, temperature, pressure, or piezo) to transmit the excitation wavelength while reflecting the Brillouin signals. The transmitted excitation wavelength light is then discarded, while the light covering the BLS peaks is coupled into the BLS spectrometer, typically via a fibre or small pinhole for spatial filtering. The alignment consists of optimally centering the filter transmission at the laser wavelength, and subsequently optimising the coupling into the output fibre or pinhole. Commercial solutions, by the company Light Machinery, based on a 4-pass pressure-tuned etalon which produces up to 70 dB suppression are available. Other types of interferometers can be used to filter out the elastically scattered light, such as modified Michelson interferometers¹³ or prism-based interferometric filters¹⁸.

While the development of integrated filters is ongoing¹⁹, a common-path birefringence-induced phase delay (BIPD) filter providing up to 60 dB extinction ratio in a single pass has been recently proposed²⁰, and made commercially available by the company *Specto Srl*. This BIPD filter can be optimized for all visible wavelengths without the need of sophisticated laser lock-in systems.

In general, when using such elastic filter(s) one should report their FSR, the type and arrangement of the filter(s), their experimentally obtained extinction, and the manufacturer(s) and model(s) of any elements employed in their realization.

Radial VIPA: Recently the concept of the VIPA has been extended by replacing the entrance *slit* of the VIPA with a *hole*, effectively changing the Cartesian degree of freedom the conventional VIPA offers (which is exploited in e.g. line-scanning BLS^{21,22}) to a radial one²³. With a suitable excitation scheme this allows one to instantaneously measure (image) the frequency shift, and thereby acoustic speed and longitudinal modulus, in different (azimuthal) directions in a sample and determine the mechanical anisotropy. Being by construction limited to a single pass/VIPA configuration, it requires good suppression of the elastic scattering (as may be achieved by e.g. absorption cells—see above), and currently still only suited to reasonably transparent samples. For dynamic measurements and measurements with low (physiologically acceptable) laser exposure it is necessary to perform angular binning to obtain statistically useful spectra. In addition to reporting the typical parameters for VIPA spectroscopy such as the FSR, and that of elastic scattering suppression schemes described above, it is also important to report the range of wavevectors/scattering-angles probed in the sample (determined by the optical setup before the spectrometer) and the azimuthal and polar angles that are being averaged over for each spectral projection.

1.2. Stimulated BLS

Stimulated Brillouin scattering (SBS) is a nonlinear phenomenon driven by the interaction between optical and acoustic fields through electrostrictive, absorptive, and photoelastic processes under energy and momentum conservation conditions. In SBS, induced density variations (i.e., an acoustic wave) in the medium are typically driven by the interference of two incident optical waves which are detuned in temporal frequency and/or spatial frequency around the BLS resonances of the medium. The stimulated acoustic wave can then be observed in the backward direction as an increase (or a decrease) in the intensity of a weak optical wave or can be interrogated in the forward direction by a third readout wave²⁴. There are currently two types of BLS microscopy that are referred to as SBS microscopy (*Frequency domain* SBS and *Impulsive* SBS), although Time-Resolved Brillouin Scattering (TRBS) also technically falls into this category (and is thus included here).

SBS microscopy's major limitation is the complexity of the experimental arrangement and the difficulty of the alignment of counterpropagating beams and scanning one of the frequencies. In its earliest implementation²⁵, the primary challenge came from the signal strength limited by the allowed pump power; however, through pulsed excitation^{26,27} and quantum-enhanced imaging²⁸ the problem of high average power lasers can be largely mitigated. Alternatively the laser exposure can be reduced significantly by only measuring at discrete frequencies²⁹.

Spectral calibration in SBS is typically done by beating measurements or locking one laser to a rubidium absorption line and using a wavemeter to measure the second (as described in

Main Text). Alternatively, one may also use a range of atomic absorption peaks (for example the four absorption lines of Rubidium 85 and 87 D2) as frequency references to calibrate the spectra, by locking the non-scanned laser to one of the hyperfine structures of the absorption peaks and adjusting the frequency scan range to cover the absorption peaks and the stimulated BLS peak. By aligning the traces measured by the lock-in amplifier with the Doppler-broadened absorption spectra measured at the DC channels, the stimulated BLS peaks can be located from the four fitted absorption peaks. The nonlinearity and offset of the frequency scanning in the standard stimulated BLS pipeline can be calibrated by comparing measurements of the same materials to those of the Rubidium absorption reference.

Frequency domain SBS

In frequency domain SBS, two counter-propagating, frequency-detuned optical fields interfere in the medium^{30,31}. One field is relatively intense (pump) and the second is typically weaker (probe). The wave interference induces travelling density variations (i.e. an acoustic wave) in the medium primarily via electrostriction. This consecutively produces a travelling refractive index variation through photoelasticity. Under proper incidence polarisation circumstances and frequency detuning that closely matches Brillouin lines of the medium (on the order of 5 GHz at near infrared wavelengths), the pump intensity is reflected from the moving grating with a Doppler shift of the probe light. As a result, the intensity of the probe increases (decreases) with a fractional gain (or loss). The corresponding lifetime of the acoustic wave is ~1 ns at near infrared wavelengths, resulting in BLS linewidths of several hundreds of MHz. When publishing SBS results it is additionally important to report the frequency range over which the probe laser is scanned and the corresponding frequency steps.

Impulsive SBS

In Impulsive Stimulated Brillouin Scattering (I-SBS), two pulses interfere to stimulate acoustic phonons within the sample. The readout process involves Bragg diffraction of a probe beam, commonly a CW laser at a different wavelength, which results in an intensity-modulated time-signal^{32,33}. Unlike above-mentioned techniques, I-SBS measures in the time domain, allowing for potentially higher time resolution, albeit at reduced spatial resolution. The excited acoustic frequency (f) in I-SBS can be calculated via $f = 2V/d$ $f = 2 \cdot v / d$ (for electrostriction), where V is the sound speed, and d is the interference fringe spacing. The frequency is tunable by adjusting the intersection angle θ of the pulse beams. The frequency resolution of I-SBS, or the ability to distinguish between two frequencies in the spectrum, is inversely proportional to the signal length.

The signal length depends predominantly on the size of the excitation volume. A high pulse energy is required to generate high SNR, but requires a good balancing with pulse length and repetition rate. When publishing SBS results, it is thus additionally important to report laser pulse characteristics. The interference fringe spacing of the two excitation beams acts as a lower limit for the spatial resolution, because the probe beam has to diffract at least one

acoustic period. One possible application of I-SBS can be *Brillouin cell cytometry*³⁴ where a typical spot size of 10 μm is employed.

Unlike spontaneous BLS the I-SBS frequency shifts are by design independent of the refractive index. This explains the notably more significant separation between the cyclohexane and water peaks seen in Fig 1E, since the refractive index of cyclohexane is $\approx 7.5\%$ larger than that of water. Since spontaneous (and Time-domain) BLS measure the frequency shift $\nu_B \propto qV \propto nV$ (where q is the scattering wavevector, V is the sound speed, and n is the refractive index), the lower (higher) sound speed in cyclohexane (water) will be countered by the higher (lower) refractive index of cyclohexane (water), such that the difference in the measured frequency shift between the two is not as significant as it would be with I-SBS.

Time resolved BLS

Time-domain Brillouin Light Scattering (TRBS), also referred to as *Phonon Microscopy* in the literature, is a collection of techniques that are generally characterised by the following two conditions: (a) there is a photoacoustic generation of coherent acoustic phonons using a pulsed pump laser and a transducer (typically a thin metallic film). (b) One measures the temporal modulation of the intensity of a second laser beam probing the propagation of the generated phonon, that is representative of the phonon time-of-flight^{35,36}.

Compared with other BLS imaging techniques, time-domain is capable of spatial resolutions down to the acoustic wavelength, and also provides access to the instantaneous phase of the acoustic wavefront. To construct a time-domain BLS optical system, it is important to consider the 3D spatial overlap of the pump and probe beams. For the initial stages of alignment, it is most convenient to use the top surface of the opto-acoustic transducer as the reference plane. The pump laser should be collimated entering the back aperture of the objective lens. Due to the incompatibility of shearing interferometers with ultrafast pulsed lasers, collimation is verified by visually inspecting the beam size at infinity while adjusting the length of the beam expanding lens pair. The position of the brightfield camera can then be optimised such that the sample and pump beam are both in focus. Next the process is repeated for the probe beam to match the focal plane of the pump and brightfield optics.

The transient reflectivity or transmissivity of the probe beam is typically detected by a ~ 5.5 MHz bandwidth amplified (10dB) photodiode, further amplified (24dB), and low pass filtered (11 MHz) before digitisation by an oscilloscope. The time signal on the oscilloscope should be maximised in amplitude by adjusting the x, y, and z overlap of the pump and probe beams. Once maximised the signal can be processed according to the protocols set out in³⁷ to extract the phonon time-of-flight, which should be normalised by the DC light level (equivalent to the amplitude of the Rayleigh peak) to achieve units of modulation depth. This processed modulation depth signal can be analysed in the frequency domain through Fast Fourier Transform to measure the BLS frequency shift ν_B . The phonon attenuation rate is related to

the longitudinal viscosity and can be measured directly in the time-domain by fitting a decaying exponential to the TRBS signal. The decay constant (α) is typically interpreted in units of μm^{-1} by converting the time-base of the signal into the axial spatial domain through the inferred relationship between the BLS frequency shift and sound velocity. However, in order to provide an equivalent BLS linewidth measurement in the frequency domain (after taking an FFT of the time-signal), the finite nature of the time window must be considered as this introduces a spectral artifact in the form of the Fourier transform of a Heaviside function (truncation of the time window). This can be calculated analytically and results in the following correction formula: $\Gamma_B \approx \Gamma_{TRBS} - 3.78/(2\pi T)$ where T is the length of the time window and Γ_{TRBS} is the full width half-maximum of the Brillouin spectral peak.

It is recommended to report the modulation depth and SNR of a given time-domain BLS setup, along with experimental parameters such as the average optical power, NA of the illuminating objective lens, laser repetition rate, and pulse width. The modulation depth scales with the acoustic amplitude (proportional to pump intensity) and probe intensity: $I_{\text{pump}}I_{\text{probe}}$. Consequently, the specimen type will dictate the sustainable pump and probe intensities (and resulting transducer heat rise). Utilising sapphire substrates and pump and probe powers of 1 and 3 mW respectively, a modulation depth of $\sim 10^{-4}$ and SNR of ~ 70 can be obtained³⁶ using a transducer with 20:160:20 nm Au:ITO:Au layers. For living cells less power should be used (0.5mW pump and 1mW probe) and typically modulation depths on the order of 10^{-6} and SNR of ~ 40 are obtained with transducer heating on the order of $\sim 1^\circ\text{C}$ ^{36,38}. To estimate the SNR of a time-domain signal, the amplitude of the BLS peak is divided by the standard deviation of amplitudes in the band of interest from an additional measurement with the pump beam blocked. The SNR will scale proportionally to the pump and probe intensities, $\text{SNR} \propto I_{\text{pump}}\sqrt{I_{\text{probe}}}$ with typical values in the range of 40-70.

Accurate measurement of the BLS frequency shift from time-resolved Brillouin scattering (TRBS) signals can be affected by artefacts generated from several signal components. When both pump and probe pulses overlap in time, optical absorption induces a strong electronic excitation within the metallic transducer and results in a sharp signal response known as the coincidence peak. This manifests itself as a rapid change in the optical properties, temperature, and ultimately the propagation of coherent phonons. The response typically relaxes according to the thermal diffusivity of the system before the arrival of the next coincidence peak. Once the sample volume has been excited mechanically there are additional frequency components in the signal which should be digitally filtered out. The mechanical resonance of the transducer ($\sim 10\text{GHz}$) in sapphire decays after only one or two cycles but it can be more prominent on softer substrates; and the BLS frequency shift of the substrate material which for stiff materials exceeds the detection bandwidth of the system can be digitally low pass filtered.

To isolate the TRBS signal from these components, the following digital operations may be performed in post-processing³⁹: cropping the signal after the coincidence peak, low-pass filtering, and subtraction of the thermal background via low-order polynomial fitting. Cropping

too close to the coincidence peak can cause reduced SNR and BLS frequency shift measurement precision due to the presence of broadband signal components, and cropping too far artificially attenuates the desired signal. Similarly, a polynomial fit order that is too low can cause errors in the determination of the BLS frequency shift, whereas an aggressive polynomial can act as a high pass filter.

The frequency resolution of the system is given by the repetition rate of the lasers (80-100MHz) which is typically insufficient to detect small variations in the BLS frequency shift in biological tissue. However, since the bandwidth of the signal is much greater than the frequency bin spacing, the spectrum can be interpolated by zero padding before calculating a Fourier transform (2^{14} - 2^{16}), allowing one to determine the centre frequency of the spectrum with ~ 10 MHz precision³⁷. In the case of estimating the longitudinal sound attenuation coefficient, it is important to note that a short optical depth of focus at high NA can lead to measurement errors. This is due to the signal decaying by loss of optical intensity rather than acoustic intensity. The acoustic attenuation in water at 5.1 GHz and at room temperature is $\sim 2.5 \times 10^5 \text{ m}^{-1}$, and comparing deviations to this value can provide a means for correction²⁹.

In time-domain BLS the photo acoustically stimulated coherent phonon field occupies a well-defined region in space, defined laterally by the PSF of the pump laser at the transducer interface, and axially by an exponentially decaying envelope set by the phonon path length, and with a well-defined wavevector (approximately a plane wave). It follows that the fundamental limit for lateral resolution is defined by the convolution of the spatially overlapped pump and probe PSF. Since the measured time of flight of the scattered probe beam is proportional to the axial distance, its period is that of the acoustic wavelength (which is shorter than the optical wavelength by a factor $\lambda_{\text{optical}}/2n$), and time-frequency analysis allows for measurements of ν_B with an axial resolution down to that of the acoustic wavelength^{38,40}. Conceivably the lateral spatial resolution can also be reduced to that of the phonon wavelength if one were to employ opto-acoustic lenses and time of flight analyses.

Some parameters important to additionally report in TRBS are the modulation depth and SNR of the TBBS setup, as well as the repetition rate and pulse width of all employed lasers.

1.3. Heterodyne BLS

Heterodyne detection is a potent technique utilised in various optical measurements, including BLS. In standard BLS detection, the measured signal is a measure of the number of photons collected within a given bandwidth and integration time, that is proportional to the square of the scattered electric field. Heterodyne detection entails mixing the scattered light with a reference beam, termed the local oscillator (LO), at a slightly different frequency. The interference between the two beams generates a beat frequency in the radiofrequency (RF) range, which is linear with the weak BLS electric field and the strong LO field. This setup often operates with LO power around 1 mW and scattered light power in the femto-Watt range, shifting the challenge from low light optics to low electric signals in the sub-GHz region.

An analog heterodyne detection setup⁴¹ typically includes several key components. A monochromatic light source passes through the sample, where it undergoes BLS. The scattered light is then mixed on a square law photodetector with the LO beam. A microwave driven electro-optic amplitude modulator generates the LO by shifting the source at specific frequencies close to the Brillouin peak. This brings the beat frequency into the sub-GHz range, where the two electric signals are amplified, filtered using a bandpass filter, and further cleaned up by a lock-in based detection. The width of the bandpass filter determines the spectral resolution of the BLS measurement.

A digital heterodyne detection setup⁴², on the other hand, incorporates high-speed digitizers and Field-Programmable Gate Arrays (FPGA) for data processing. Similar to the analog setup, the scattered light and EOM generated LO beam are mixed, producing the beat frequency. The digitizer samples the beat signal at high rates, such as 3.2 Gs/s, and the FPGA processes the data in real-time. This digital approach offers higher processing speed and flexibility, enabling real-time data analysis.

Heterodyne detection is an intrinsically single mode measurement; thus the usual single mode fibre collection of Brillouin microscopy confocal layout is best suited for this approach. The strength of this approach is that, thanks to the down conversion of the optical heterodyne mixing, contrast and resolution are no longer a major issue, as they are not limited by the transfer function of optical devices, but benefit from the tools and precision offered by electronics in the sub GHz region. As a consequence, the approach promises a calibration free, extremely stable and compact layout.

When using a heterodyne detection setup it is additionally important to report the manufacturer(s) and model(s) of instruments used, the achieved spectral resolution, as well as characteristics of the bandpass filter and when relevant those of the Analog to Digital Converter (ADC).

1.4. Fibre probes

BLS fibre probes are devices made of or compatible with optical fibre components that can replace the confocal microscope (“front-end”) of a BLS imaging system. Different designs have been proposed and demonstrated to date based on whether signal acquisition occurs in the frequency or time domains.

For the frequency domain, BLS fibre probes can be constructed using a single or dual optical fibre configuration and then supplemented by a standard VIPA or a scanning TFP spectrometer to produce a fibre-integrated BLS measurement setup. The main drawback in using optical fibres for delivery of the laser light and for the collection of the backscattered light, is the spontaneous BLS that occurs inside the fibre core. Since light is tightly focused inside the core of the optical fibre, and a typical length of an optical fibre is a few metres, the

effective interaction volume between light and fibre core material is significantly greater than the volume of the optical voxel formed by the focussed beam in the sample. Thus, despite the fibre BLS frequency shift usually being significantly larger than that of the probed sample (>20 GHz), this parasitic signal might present a challenge in detecting the BLS signal originating from the sample³⁵. Different solutions to this problem exist, including: 1) hollow-core optical fibres for delivery of the laser illumination to the sample without inducing any significant Brillouin scattering in the fibre itself⁴³; 2) the use of dual-core or dual-fibre geometries, as is done in Raman endoscopy, where delivery and collection channels are separated thus avoiding mixing of the fibre backscattering signal with that from the sample⁴⁴. Dual fibre approach requires careful design of the focusing optics in order to ensure collection of the inelastically scattered light back into the core of the second fibre. This can be done by creating on demand 3 dimensional microoptics and/or compound lenses fabricated via two-photon-polymerisation laser writing technique. Overall, the imaging resolution in lateral and axial directions depend on the focusing micro-optics design and is typically slightly worse when compared to conventional microscopy objective lenses, owing to low numerical aperture of the single-mode optical fibre and fibre-compatible lenses⁴³.

Brillouin fibre probes can also be constructed to operate on the principles of TRBS; however this requires fabrication of an optoacoustic transducer onto the fibre tip for the generation of coherent phonons. These stimulated phonons provide an enhanced BLS cross-section, however currently at the cost of slower detection electronics. The time-domain BLS process is largely insensitive to BLS back scattering along the glass fibre - since the high amplitude stimulated phonons are localised to within 10-20 μm of the transducer as is the coherence length of the interaction. This means that standard silica fibres of arbitrary length can be used.

A time-domain BLS system can be converted for fibre implementation by fibre coupling the free-space pump and probe beams and coupling these into a common channel via a 2:1 fibre coupler. The common channel contains an inline fibre circulator, containing the sample-facing fibre at one port and the detection photodiode at the other, thus avoiding losses incurred by using standard 3dB couplers. If the pump and probe beams contain separate wavelengths, then SNR can be further improved by using a wavelength division multiplexer to attenuate the pump beam at the detector and reduce its contribution to shot noise³⁷. SNR and modulation depths will be comparable to free-space TRBS provided similar parameters are used for the optical system and optoacoustic transducer.

Time-domain fibre probes have been shown can achieve 2 μm lateral resolution, realised by point-scanning the sample or fibre, and down to 260 nm depth resolution (equivalent to the phonon wavelength) which is achieved without confocal scanning the optics since the time-domain signal encodes the depth domain⁴⁰. However, high depth resolution comes at the cost of reduced depth measurement range (~ 10 μm) which can be increased by lengthening the probe wavelength (the attainable depth scales as the square of the probe wavelength).

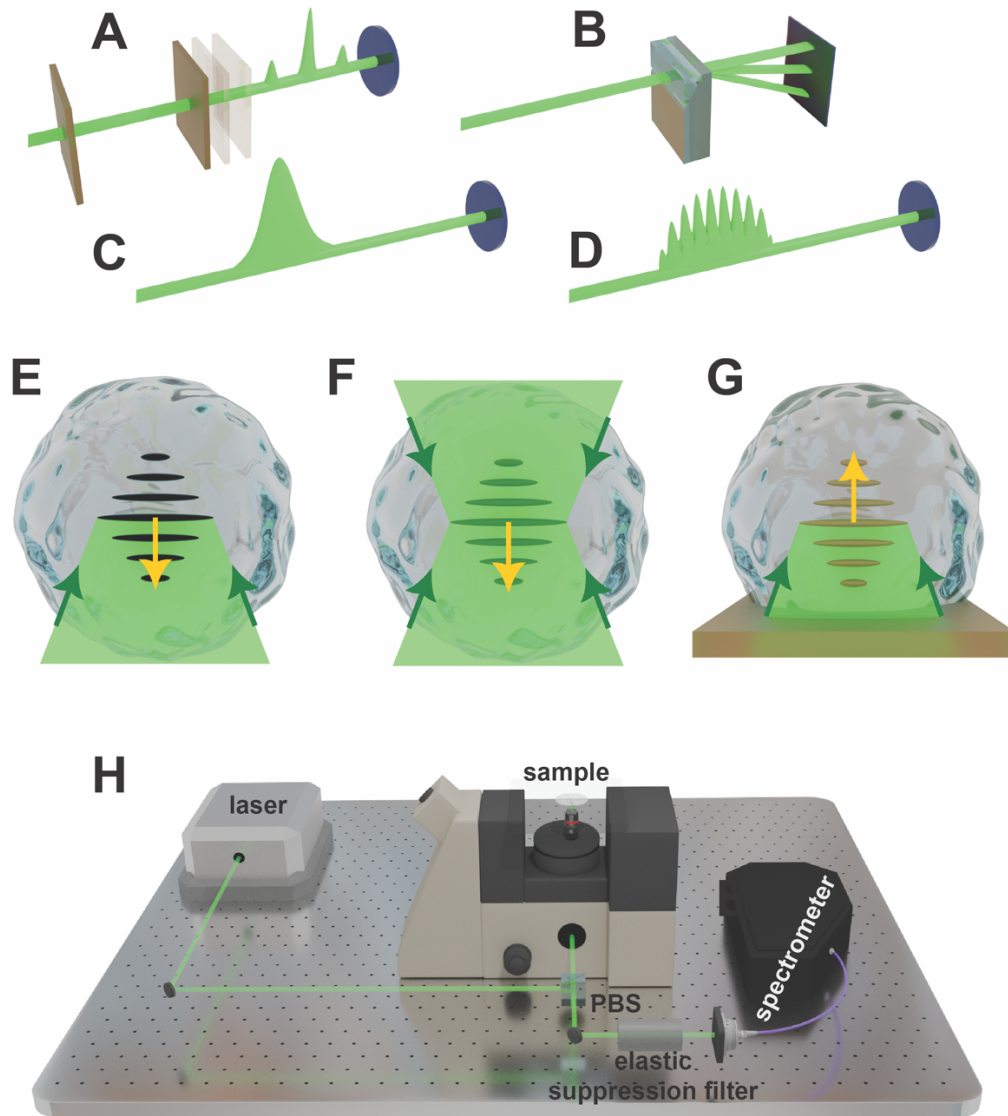
The exact details of the fibre inelastic scattering depend on the type of fibre used, its length, the amount of bending of the fibre, and other properties of the experimental configuration.

Even the use of hollow-core optical fibres does not remove the fibre background completely since a small percentage of the fundamental fibre mode and higher order modes interact with the microstructured region of the hollow-core fibre, producing a small amount of backwards-scattering BLS⁴³. Such a parasitic signal, however, does not interfere with the desired BLS signal from the biological sample, as it is positioned outside the spectral region of interest, and can be removed during data post-processing. High order fibre modes may also be attenuated by bending the hollow-core fibre slightly, without inducing significant bending losses to the fundamental fibre mode.

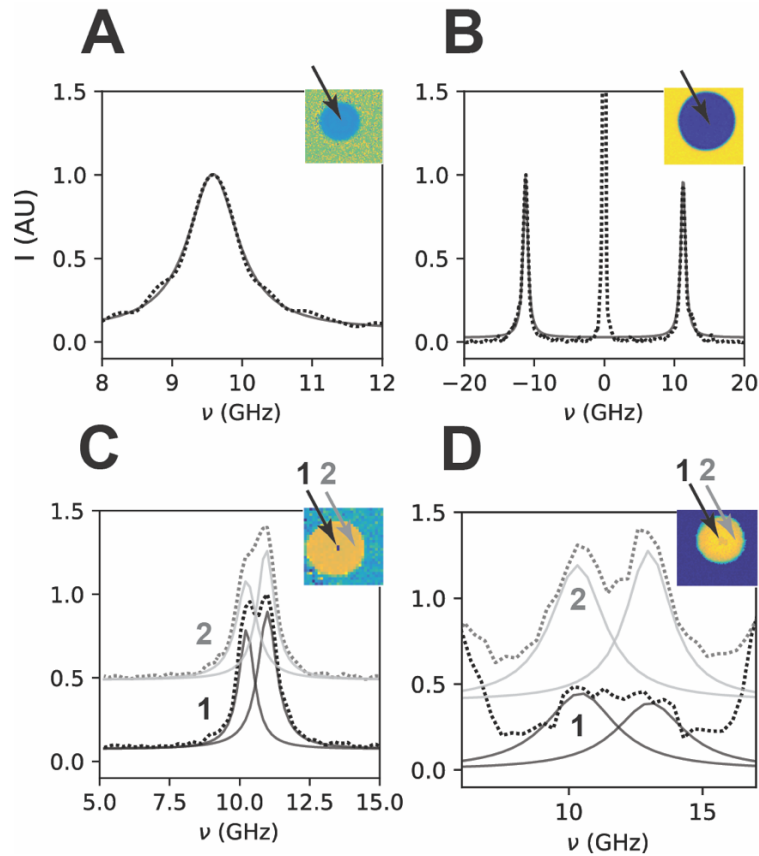
As with spontaneous BLS fibre probes, in TRBS fibre probes, the BLS signal from the glass fibre is typically low pass filtered in post-processing. It is worth noting that the time-domain BLS process is only sensitive to the stimulated coherent phonons within the vicinity of the transducer, and although these are approximately 6 orders of magnitude greater amplitude than the spontaneous thermal phonons along the fibre length, they attenuate very rapidly, travelling away from the transducer within tens of microns. This allows one to use long lengths of glass fibre without adversely affecting the glass signal amplitude. Additionally, the acoustic impedance and photoelastic coefficient mismatches between the glass, metal and water layers result in a ~2 orders of magnitude greater signal amplitude from the water or cell tissue interface compared to that from the glass interface. Sample heating should be considered when performing fibre-based time-domain BLS both in terms of specimen viability and changing the measured BLS frequency shift. The former can be mitigated through lowering optical fluence and the inclusion of thermally conductive transducer layers. Any thermal gradient produced by the fibre probe can be calibrated by obtaining reference measurements (away from a specimen) in the aqueous couplant medium. These have been found to be stable under normal experimental conditions, such that subtracting these frequency offsets from the specimen BLS shifts can potentially compensate for such thermal contributions^{37,40}.

When using fibre probes it is additionally important to report the manufacturer and model of the optical fibre, any strategies used to overcome the contribution from BLS backscattered light from the fibre, the coupling efficiency of reflected and/or Brillouin scattered light back into the fibre-core prior to detection, and as relevant any treatment performed on the tip of the fibre. When using transducers at the end of fibre probes, it is additionally desirable to present an assessment of the sample heating induced by the transducer, and any relevant calibrations made for this.

The different BLS measurement schemes and different probing-detection geometries are illustrated in Supplementary Figure 1. Supplementary Figure 2 shows characteristic BLS spectra of a polystyrene bead measured using spectrometers in different laboratories (corresponding to the spatial maps shown in Figure 2 of the Main Text).



Supplementary Figure 1: Brillouin Light Scattering Spectrometer measurement schemes. A-D: Spectral acquisition modalities for different spectrometers: (A) *Scanning Fabry Perot*. Etalon(s) are spatially scanned to sequentially read out different wavelengths on a single pixel detector. (B) *Virtual Imaged Phase Array (VIPA)*. light is angularly dispersed and imaged on a detector array to render a spatially resolved projection of the spectra. (C) *Stimulated Brillouin Scattering*. The wavelength of a probing laser is tuned, and spectra is read out sequentially on a single pixel detector. (D) *Time-domain Brillouin Scattering*. The delay time of a probing laser pulse is tuned, and delay is read out sequentially on a single pixel detector. **E-G:** Different probing and detection geometries used in BLS microscopy. Incident probing light is shown in green, and the direction from which scattered light is detected is shown by yellow arrows (E) Back scattering geometry typically used in spontaneous BLS microscopy. (F) Bi-directional illumination used to generate transient grating in Stimulated Brillouin scattering. (G) Signal detection from above (scattered from transducer generate acoustic modes, excited from below). **H:** Schematic showing key components of frequently employed setups used for spontaneous BLS microscopy of biological samples, showing single mode continuous wave laser source, elastic suppression filter and imaging spectrometer (see also Supplementary Text, section 1.1).



Supplementary Figure 2: Brillouin Light Scattering spectra of polystyrene beads. Representative spectra from indicated positions of the spatial maps presented in Figure 2 of the Main Text. The panels **A-D** correspond to the respective spatial maps in Fig 2 (A-D). Insets show spatial maps of BLS frequency shift (viz. Fig 2) with arrows indicating the positions on the map from which the plotted spectra are shown.

2: Some useful equations for BLS microscopy

The interpretation of Brillouin Light Scattering (BLS) measured parameters is historically divided between formalisms that consider the measured parameters in two different frameworks.

Firstly, there is that of acoustic (e.g. shock) waves moving through a solid and largely elastic material, which is relevant for describing e.g. geological activity, mechanical properties of polycrystalline materials, etc... Here the material properties are often treated as effective entities, with the interest typically being how the collective, and for the most part elastic, properties change e.g. in different directions. Given the characteristic relaxation times of *hard* solids are very slow, the BLS derived *stiffness coefficients* can be useful for modeling also lower frequency acoustic and mechanical properties.

Secondly, there is the thermodynamic and (molecular) hydrodynamic picture, which considers the molecular interactions on short time scales to predict the supported phonon

modes. From this one can gain insight into the electrostatic screening of polymers (such as DNA) and structural states of molecules that affect their interactions with a solvent or molecules in their immediate vicinity. Here the time scales probed using BLS may become more comparable to those of the relaxation processes, and one can gain insight into the nature of e.g. phase and structural transitions by virtue of the changes in the hypersonic speed and attenuation. The interpretation here is less on our everyday understanding of viscoelastic moduli, but rather on elucidating changes in “viscoelastic properties” on time scales that affect the molecular interactions, namely relaxation times, and how these may lead to changes in both molecular processes and collective material properties.

Biological systems, such as a living cell, rarely fit either of these two, but rather often exist in a somewhat undefined dynamic state between an anisotropic solid and a liquid, usually containing elements of each. In regard to building on the established BLS nomenclature this presents a unique challenge. Below we define some parameters, taken from these two established branches, that can prove useful for analysing BLS measurements in biological systems. These include both basic fitting parameters and biophysically relevant parameters, with a brief justification of their relevance in each case. These are intended to serve as a potentially useful, but in no way complete, reference for people performing BLS on bio-relevant matter, to assure that all parameters are defined consistently.

2.1. Fitting equations

Damped Harmonic Oscillator: In the frequency domain, the intensity of each BLS peak is functionally described by a Damped Harmonic Oscillator (DHO) given by:

$$I(\nu) \propto \frac{\Gamma_B \nu_B^2}{(\nu^2 - \nu_B^2)^2 + (\Gamma_B \nu_B)^2} \quad \text{Eqn. S1}$$

where ν is the frequency shift relative to the probing laser frequency, and ν_B and Γ_B are the BLS frequency shift and BLS linewidth. If ν_B falls in the vicinity of a structural relaxation processes ($\sim 1/\tau$, where τ is the viscous relaxation time) the shape of the peak will generally have a more complex form, and its spectral shape needs to be calculated from first principles^{45,46}.

Lorentzian Function: To an, often good, approximation the BLS peak can also be described by a Lorentzian function:

$$I(\nu) \propto \frac{\Gamma_B^{(L)}}{(\nu - \nu_B^{(L)})^2 + (\Gamma_B^{(L)}/2)^2} \quad \text{Eqn. S2}$$

where $\nu_B^{(L)}$ and $\Gamma_B^{(L)}$ are the (Lorentzian) BLS frequency shift and BLS linewidth. If the linewidth is not too large this is usually a good approximation.

Correction to Lorentzian fit: Due to the general asymmetry of the DHO, the Lorentzian fitting parameter $\nu_B^{(L)}$ will differ from the true (DHO obtained) ν_B by an amount that becomes more significant with increasing linewidth given by:

$$\nu_B = \nu_B^{(L)} \sqrt{[1 + (1/2)\Gamma_B^{(L)}/\nu_B^{(L)}]} \quad \text{Eqn. S3}$$

2.2. Frequency shift

The frequency shift ν_B of a given BLS peak can be expressed in terms of the hypersonic speed V_q of the respective measured phonon modes in the direction of the scattering wavevector q :

$$\nu_B = (2\pi)^{-1}\omega_B = \pm(2\pi)^{-1}qV_q \quad \text{Eqn. S4}$$

The scattering wavevector q (which is also the wavevector of the created/annihilated phonons) is given by:

$$q = 4\pi n \sin(\theta/2) / \lambda_0 \quad \text{Eqn. S5}$$

Here θ is the scattering angle (=180 degrees in the back-scattering geometry), n is the refractive index, and λ_0 is the free-space wavelength of the probing laser.

In the vast majority of BLS microscopy studies on biological samples (and to date practically all studies on living cells) one measures only the *longitudinal acoustic phonons*. These in almost all cases have larger BLS frequency shifts than the transverse phonons also supported in solids and some liquids, and a much larger scattering cross section. They correspond to travelling density waves where the displacement of molecules is entirely in the direction of the propagating waves. Namely, solutions of the wave equation:

$$\frac{\partial^2 u_i}{\partial t^2} = V^2 \frac{\partial}{\partial x_j} \left(\frac{\partial u_l}{\partial x_k} \right) \quad \text{Eqn. S6}$$

in which u_i is the displacement of molecules in the direction x_i , and t is time, with $i = j = k = l$. The speed V here is related to the mass density (ρ) of the material, as well as its stiffness. The relevant stiffness here depends on the direction of the strains relative to the stresses. These can be expressed in terms of the generalized Hooke's Law:

$$\sigma_{ij} = c_{ijkl} \varepsilon_{kl} \quad \text{Eqn. S7}$$

Where σ_{ij} and ε_{kl} are the strain and stress tensor (and the subscripts indicate the decomposed orthogonal vector directions, i.e. $i = 1,2,3$). c_{ijkl} is the so-called *stiffness tensor* and describes the complete elastic response of a material in an arbitrary direction subject to a stress in the same or a different arbitrary direction. c_{ijkl} is a huge tensor, but owing to several symmetry constraints only has 21 independent components and is usually more compactly

expressed as a 6 x 6 tensor c_{ij} . The first three diagonal components (c_{11} , c_{22} and c_{33}) will correspond to longitudinal modes (i.e. where $i = j = k = l$) in the three orthogonal directions.

In general the relation between the hypersonic speed in the direction of the probed scattering vector $\hat{q} = \mathbf{q}/|\mathbf{q}|$, and the relevant stiffness tensor components can readily be obtained from the Christoffel equation:

$$|(\hat{x}_k \cdot \hat{q})(\hat{x}_l \cdot \hat{q})c_{ijkl} - \rho V_q^2| = 0 \quad \text{Eqn. S8}$$

where $\hat{x}_i$ are unit vectors in the direction of the respective subscripts. It follows that the speed of the longitudinal phonons will in the direction $\hat{x}_i$ be given by:

$$V_i = \sqrt{c_{ii}/\rho} \quad \text{Eqn. S9}$$

where c_{ii} describes the response to a stress and strain in the same direction (i).

The longitudinal elastic modulus will normally correspond to the stiffness tensor components c_{11} , c_{22} or c_{33} when the material is measured in a symmetry direction such as along a symmetry axis for a crystal, along the fibre axis for a fibre, or in essentially any direction for an isotropic material like an amorphous material, gel, or liquid. Conceptually the longitudinal elastic modulus can be thought of as the relevant modulus when the strain is uniaxial. This means that the material is stretched or compressed only in a single direction and does not change in the perpendicular directions.

If one assumes that the material is mechanically isotropic (has same elastic modulus in all directions) then $c_{11} = c_{22} = c_{33} = M'$ and it follows from Eqn. S4, S5 and S9 that:

$$\nu_B = 2n\lambda_0^{-1} \sqrt{\frac{M'}{\rho}} \sin\left(\frac{\theta}{2}\right) \quad \text{Eqn. S10}$$

which is the key equation used to calculate the longitudinal elastic modulus from BLS frequency shift measurements. It is important to note that this is for a single scattering angle (θ), whereas one in practice measures over a finite range of scattering angles (on account of the finite NA of the microscope objective), which will result not only in peak broadening but also potential shifts in ν_B . To this end one should ideally integrate Eqn. 1 over the solid probing and collection angles, especially for scattering geometries deviating from back scattering, to obtain accurate measures of M' .

The above assumes that not only the longitudinal elastic modulus but also that the refractive index (n) is isotropic. The latter may not always be the case, e.g. in fibrillar structures such as muscles or cellulose. In such cases its directional dependence needs to explicitly be accounted for, which may be done by writing it in the form of a tensor ($n \rightarrow \mathbf{n}$) and considering its projection in the direction of the scattering wavevector. The BLS frequency shift can then be compactly expressed as:

$$v_B = \pm 2 \mathbf{n} \cdot \hat{\mathbf{q}} \lambda_0^{-1} V_q \quad \text{Eqn. S11}$$

Where V_q in a mechanically anisotropic sample is in general a combination of stiffness tensor components defined by the projection of the scattering wavevector that can be calculated from Eqn. S8.

It is also possible to measure BLS scattering from transverse acoustic phonons. These show up as peaks at (usually) lower frequency shifts than the longitudinal phonon peaks and typically require the implementation of distinct (non-backscattering) measurement geometries. From these it is possible to calculate the *shear modulus* (G) (also called the *transverse modulus*), with the real part (G') obtainable from the peak frequency shift in an analogous manner to the longitudinal elastic modulus using Eqn. S4, S5, S9 and S10. The shear modulus is a stiffness coefficient (c_{ij}) that describes cases where the strain is pure shear. This means that the material's shape is changed but its density is not. It is the appropriate modulus for transverse sound waves. As with the longitudinal modulus, it can be different in different directions unless the material is isotropic. It will normally correspond to c_{44} , c_{55} , or c_{66} of the stiffness tensor when measuring in a symmetry direction in a crystalline material. Typically, two of these three coefficients show up as two distinct BLS peaks in a single measurement. These two peaks correspond to the two principal polarizations of the transverse phonon mode. The transverse modes are usually measured in distinct (90 degree) scattering geometries, although in anisotropic systems they can also be revealed in back-scattering measurements, provided that the wavevector is not along a direction of high symmetry. Because they are generally isotropic, amorphous materials will only have a single shear modulus.

2.3. Linewidth

The BLS linewidth (Γ_B), i.e. the Full Width at Half Maximum (FWHM) of the BLS peak, obtained from fitting a DHO or Lorentzian function (Eqns. S1 & S2), is fundamentally determined by the lifetime of the phonon--in an analogous manner to the peak width in a high energy physics experiment, with the (quasi-)particles in this case being the acoustic phonons. Given the phonons propagate at a presumed speed V_q , this lifetime is directly related to their spatial attenuation length. It is possible to also calculate a corresponding *loss modulus* that describes the dissipative properties of the sample (i.e. the corresponding imaginary part of the stiffness tensor components). In the case of longitudinal phonon modes this would be:

$$M'' = 2\pi\rho q^{-2}v_B\Gamma_B \quad \text{Eqn. S12}$$

For a mechanically isotropic sample, M'' is the imaginary part of the *complex longitudinal modulus*: $M = M' + iM''$, and $i = \sqrt{-1}$. But as with the real part, M'' can in general take on different values in different directions (i.e. M''_{ij}). An analogous relation may be obtained from

the BLS frequency shift and linewidth of the transverse acoustic modes, yielding the imaginary part of the shear modulus (G'').

For fluids it may sometimes also be desirable to express the dissipative properties in terms of a viscosity. For this the following relation may be used:

$$\Gamma_B = \frac{q^2}{2\pi\rho} \left[\frac{4}{3}\eta_S + \eta_B + \frac{\kappa}{C_P}(\gamma - 1) \right] \quad \text{Eqn. S13}$$

Here η_S is the *dynamic shear viscosity* common to rheology (but in this case that the probed MHz-GHz frequencies). η_B is the *dynamic bulk viscosity* (also referred to as the second viscosity by e.g. Landau and Lifshitz), which is a distinct material property to the shear viscosity, that in practice can usually be neglected from fluid dynamics calculations, i.e. Navier Stokes equation, on presumption of the incompressibility of flow. It however becomes relevant when there are e.g. bubbles, one considers microscopic instabilities, or of course for the propagation of longitudinal sound waves. κ is the thermal conductivity of the sample, and $\gamma = C_P/C_V$ where C_P (C_V) are its specific heat under constant pressure (volume). In water-based matter one often can assume that $C_P \approx C_V$ such that $\gamma \approx 1$ and that the linewidth is directly proportional to the combination of the dynamic shear and bulk viscosity with no offset:

$$\Gamma_B \approx \frac{q^2\eta_L}{2\pi\rho} \quad \text{Eqn. S14}$$

Where $\eta_L = (4/3)\eta_S + \eta_B$ is the *dynamic longitudinal viscosity* (sometimes also called the *effective* or *dilational* viscosity in the literature) that is relevant for describing the attenuation of longitudinal acoustic modes. η_L can serve as a good reporting parameter insofar that in a Newtonian fluid under hydrodynamic conditions it will be independent of frequency and wavevector (compared to the loss modulus and acoustic attenuation, which would scale linearly and quadratically respectively with the probing wavevector--and thus also the refractive index). For some applications it may also be relevant to calculate the *kinematic longitudinal viscosity* μ_L , which is defined as $\mu_L = \eta_L/\rho$.

For measurements of transverse phonons, the linewidth can analogously yield information on the imaginary part of the shear modulus (G'') and shear viscosity (Eqn. S12 and Eqn. S14 with the replacement $\eta_L \rightarrow \eta_S$ in the latter). The shear viscosity and moduli measured using BLS will be distinct from that measured using classical rheological techniques (shear rheology, etc. which measure at low shear rates) as it is typically at many orders of magnitude higher frequencies.

The above assumes that there are no extrinsic factors that cause broadening of the BLS peak (i.e. increases in the linewidth) and are defined for measurements at a single scattering wavevector. In practice numerous factors will also cause broadening of the BLS scattering peak and over estimations of the linewidth. These include measurements from a distribution of scattering angles (finite NA), material heterogeneities in the probing volume (that are larger than the characteristic length scales of the probed phonons), multiple scattering (most

relevant for non-transparent samples or deep tissue imaging), and ultimately the usually non-negligible spectral Instrument Response Function (IRF) of the spectrometer. In addition, when the probed phonon frequency is in the vicinity of a structural relaxation process with characteristic time(s) τ , namely $\nu_B \sim \tau^{-1}$, as may happen to be the case in soft-matter/gels, the functional dependence of the BLS scattering peak will undergo significant broadening and deviate from a simple DHO.

While a spectral deconvolution may suffice in correcting for the IRF and finite-NA broadening, accounting for the other factors analytically or even numerically is less trivial, requiring a priori information on the sample that is usually not available. To this end a experiment/sample-specific semi-quantitative (or at least qualitative) assessment of their expected relative contributions should be made, and caution should be exercised when presenting absolute values of linewidth-derived parameters.

2.4. Loss tangent

This can serve as a useful metric of changes in the viscoelastic properties owing to it being independent of changes in the mass density (which are challenging to measure on the microscopic scale in complex biological samples). It is defined, for the longitudinal modes, as:

$$\tan \delta = \Gamma_B / \nu_B = M'' / M' \quad \text{Eqn. S15}$$

An analogous quantity may also be defined for the transverse (shear) modes. It is sometimes stated that $\tan \delta$ is independent of the refractive index n . This is only true if both M'' and M' are assumed to be independent of the phonon wavevector probed q (Eqn. S5) (or have the exact same functional dependence thereon). While potentially valid in hard dehydrated solids, this is not the case in the hydrodynamic picture, where $\Gamma_B \propto q^2$ (Eqn. S13) and $\nu_B \propto q$ (Eqn. S4), such that $\tan \delta \propto q \propto n$. In the vicinity of structural and phase transitions this assumption can be assumed to break down even more dramatically.

2.5. Effect of Acoustic Heterogeneities

Most biological materials have spatial variations in their elastic properties and this heterogeneity affects the extension of the acoustic excitations, which are probed by interaction with light in Brillouin scattering. In fact, the degree to which acoustic excitations remain confined within regions of differing stiffness is governed by the mismatch of acoustic impedance (Z), defined as $Z = \rho V$ where ρ is the mass density and V the sound velocity. This mismatch influences the partial reflection and transmission of acoustic waves at internal interfaces, effectively modulating the coupling between adjacent regions [41]. Therefore, the Brillouin spectrum acquired from a heterogeneous medium reflects, in general, the mechanical properties of the various regions contained within the scattering volume. However, its specific spectral shape is strongly influenced not only by the elastic moduli of

the individual components but also by the morphological organization and spatial distribution of the heterogeneities.

When the characteristic dimensions of the heterogeneities are significantly smaller than the wavelength of the probed acoustic phonons, or when the acoustic impedance mismatch between adjacent domains is weak, their presence does not produce resolvable features in the BLS spectrum. Here one may assume an effective material response. Under these conditions, a single acoustic mode can extend across multiple domains, effectively averaging the elastic properties on the scale of the phonon coherence length, resulting in a single homogeneous-like spectral peak. Nonetheless, the presence of sub-coherence length heterogeneities contributes to additional phonon attenuation through static scattering processes, which are superimposed on the intrinsic (dynamic) viscous damping. This manifests itself by an increase in the spectral linewidth, representing a reduced phonon coherence length⁴⁷.

When the size of the heterogeneities becomes comparable to the acoustic wavelength, and the acoustic impedance contrast between adjacent regions is sufficiently large, acoustic modes may become spatially confined within the individual domains. In this regime, the BLS spectrum no longer exhibits a single feature but rather a discrete set of peaks, each corresponding to quantized vibrational modes. The frequency positions of these peaks encode information on the local elastic properties, but their interpretation requires prior knowledge of the characteristic size and geometry of the confined structures, as the discrete nature of the spectrum will be dictated by the boundary conditions. Furthermore, the relative intensity of each mode in the Brillouin spectrum will depend upon the coupling between the acoustic field and the optical probe. A specific analysis procedure is typically required⁴⁸⁻⁵⁰. As an example, in thin films, the acoustic normal modes form a discrete series of standing waves, characterized by the wavevector

$$K_{\perp} = \frac{(2m - 1)\pi}{2d}$$

where d is the film thickness and m is an integer.

When the size of the acoustic heterogeneities is larger than the phonon coherence length, one will have distinct peaks associated with the different materials that can be analyzed independently⁵¹ (with an intensity proportional to the product of the filling factor and scattering efficiency⁵²).

2.6. Relation to other moduli

In an isotropic material, the longitudinal elastic modulus is related to the shear and bulk elastic modulus via:

$$M' = (4/3)G' + K' \quad \text{Eqn. S16}$$

This is true also for the imaginary component (loss moduli), with the longitudinal viscosity being defined analogously: $\eta_L = (4/3)\eta_S + \eta_B$ (see Eqn. S13 and S14).

In a linear, isotropic material the longitudinal modulus (M), Young's modulus (E), Shear Modulus (G) and bulk modulus (K) are related via the Poisson ratio (σ):

$$M = \frac{E(1-\sigma)}{(1+\sigma)(1-2\sigma)}; \quad E = \frac{9KG}{(3K+G)}; \quad G = \frac{E}{2(1+\sigma)}; \quad K = \frac{E}{3(1-2\sigma)}$$

$$M = \frac{1-\sigma}{(1+\sigma)(1-2\sigma)} E = \frac{2(1-\sigma)}{(1-2\sigma)} G = \frac{3(1-\sigma)}{1+\sigma} K \quad \text{Eqn. S16}$$

These equations are only valid $-1 < \sigma < 0.5$, with the material becoming inherently unstable outside of this range. In practice, their relevance will also already break down when E/G and M/K become significantly different as will be the case when σ approaches 0.5. Here one would need to know σ to an increasingly higher accuracy (to an infinite accuracy in the limiting case of $\sigma = 0.5$). It should be kept in mind that the Poisson ratio is also a frequency dependent quantity, and cannot generally be assumed to have the same values at MHz-GHz as it does at low frequencies.

While Eqn. S15 and S16 may be used to convert between moduli (assuming one has knowledge of the Poisson's ratio at the relevant MHz-GHz frequencies), the obtained e.g. E or G can in the general case not simply be compared to measurements obtained from techniques probing at quasi-static frequencies (AFM, shear rheology, etc.) due to the many orders of magnitude differences in frequency, and often unknown mechanical relaxation spectrum between these frequencies. For specific sample types it may however be possible to realize significant and useful empirical correlations between the different moduli, and even the same or different moduli at different frequencies.

An additional caveat for calculating e.g. E from the BLS measured M and G that may become relevant, is that the BLS measured M and G may in themselves be at quite different frequencies--namely the BLS frequency shifts measured for the longitudinal and transverse phonons may be a couple of orders of magnitude apart. Eqn.s S15 and S16 on the other hand assume that all quantities are at the same frequency.

Much of the above is based upon the assumption of hydrodynamic behaviour and that there are no significant phase or structural transitions at or near the probed BLS frequencies. This may or may not be the case in biorelevant matter, and this should always be kept in mind.

Finally, a difference in the BLS measurements and perturbation-based measurements of these these moduli may also be due to the latter usually measuring elastic moduli by applying a finite-strain (obtaining the so-called *engineering* stress-strain), whereas BLS probing the isentropic *true* stress-strain.

2.7. Calculation of stiffness tensor

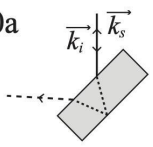
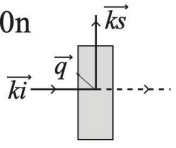
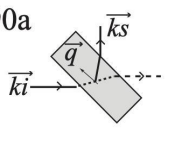
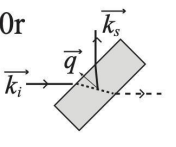
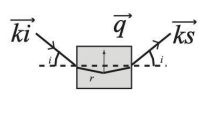
It is in principle possible to determine the complete stiffness tensor by performing BLS measurements of the sample with different scattering wavevectors. Measurements of an anisotropic sample in an arbitrary direction that does not correspond to a symmetry axis, will yield a possibly complex algebraic combination of stiffness tensor coefficients (which can be determined by Eqn. S8). Such measurements are relatively uncommon except in cases where the experimenter is explicitly trying to extract the values of all stiffness tensor elements including off-diagonal elements such as c_{12} . Most measurements are usually performed in high symmetry directions, where the effective stiffness corresponds to just a single coefficient from the stiffness tensor, such as c_{11} , c_{22} , or c_{33} for a longitudinal mode or c_{44} , c_{55} , or c_{66} for a transverse mode (see Supplementary Figure 3).

$$\begin{array}{c}
 \begin{pmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{yz} \\ \sigma_{zx} \\ \sigma_{xy} \end{pmatrix} = \begin{array}{c} \text{longitudinal} \\ \begin{pmatrix} \textcolor{blue}{c}_{11} & c_{12} & c_{13} \\ c_{21} & \textcolor{blue}{c}_{22} & c_{23} \\ c_{31} & c_{32} & \textcolor{blue}{c}_{33} \\ \hline c_{41} & c_{42} & c_{43} \\ c_{51} & c_{52} & c_{53} \\ c_{61} & c_{62} & c_{63} \end{pmatrix} \begin{array}{c} c_{14} & c_{15} & c_{16} \\ c_{24} & c_{25} & c_{26} \\ c_{34} & c_{35} & c_{36} \\ \hline \textcolor{red}{c}_{44} & c_{45} & c_{46} \\ c_{54} & \textcolor{red}{c}_{55} & c_{56} \\ c_{64} & c_{65} & \textcolor{red}{c}_{66} \end{array} \end{array} \begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \varepsilon_{yz} \\ \varepsilon_{zx} \\ \varepsilon_{xy} \end{pmatrix} \begin{array}{c} \text{longitudinal} \\ \text{shear} \end{array}
 \end{array}$$

stress shear strain

Supplementary Figure 3: Stiffness Tensor. BLS in a symmetric direction measures the blue (longitudinal modes) or red (transverse modes). All other stiffnesses are of mixed quasi-transverse / quasi-longitudinal nature and are found by fitting the BLS signal to the Christoffel Equations.

To measure all the stiffness tensor elements requires measuring the sound speeds along different directions which will correspond to the different stiffnesses (c_{11} , c_{22} , etc.) using a combination of different scattering geometries and rotations of the sample. Some common scattering geometries are shown in Supplementary Figure 4. The BLS experiment should be altered depending on the symmetry of the sample. A truly isotropic material only has two independent stiffness coefficients (c_{11} the longitudinal component, and c_{44} the transverse component) and these are easily measured, without needing to know the refractive index, by using an equal angle scattering geometry (90a, see Supplementary Figure 4). This can however be complicated in cells and other small materials. A single spectrum in an equal angle geometry will give a longitudinal mode, which gives c_{11} , and a transverse mode, which gives c_{44} . This gives all the stiffness tensor elements for an isotropic material.

Geometry	Schematic	Equation
180	180a 	$V_{180} = \frac{\Delta v_{180} \lambda_o}{2n}$
90n	90n 	$V_{90n} = \frac{\Delta v_{90n} \lambda_o}{n\sqrt{2}}$
90a Equal-Angle	90a 	$V_{90a} = \frac{\Delta v_{90a} \lambda_o}{\sqrt{2}}$
90r	90r 	$V_{90r} = \frac{\Delta v_{90r} \lambda_o}{\sqrt{4n^2 - 2}}$
Platelet (60°)	Platelet 	$V_{60} = \Delta v_{60} \lambda_o$

Supplementary Figure 4: BLS Scattering Geometries. Some common BLS scattering geometries used for determining the diagonal stiffness tensor coefficients in materials of known symmetries. $k_i(k_s)$ are incident and scattering wavevector, q is the scattering wavevector, and Δv is the measured frequency shift.

For some applications, one wants to know other elastic properties such as the Young's modulus (E), the bulk modulus (K), the Poisson's ratio (σ), or the shear modulus (G), averaged over crystal orientations. These can all be derived from the stiffness coefficients c_{ij} . For an isotropic material we have three distinct stiffness coefficients c_{11} , c_{12} , and c_{44} , but only two are independent because of the constraint $c_{44} = (c_{11} - c_{12})/2$. The different moduli can then be found from:

$$E = \frac{3c_{12}c_{44} + 2c_{44}^2}{c_{12} + c_{44}} = \frac{(c_{11} - c_{12})(c_{11} + 2c_{12})}{c_{11} + c_{12}}$$

$$G = c_{44}$$

$$K = \frac{1}{3}(c_{11} + 2c_{12})$$

$$\sigma = \frac{c_{12}}{c_{11} + c_{12}} \quad \text{Eqn. S17}$$

As also apparent from Eqn. S17, these equations show that the longitudinal modulus c_{11} is not the same as the Young's modulus but is some combination of the Young's modulus and the Poisson's ratio – expressed in this case as combinations of stiffness tensor elements.

Determining the entire stiffness tensor in a more complex anisotropic sample requires measurements in several directions of the material including off-axis or off-symmetry directions. The experimental design must ensure that the measurements are sensitive to all the stiffness tensor elements and not just the ones along the main diagonal. This means measuring not just pure longitudinal and transverse modes but also mixed modes, so-called *quasi-longitudinal* and *quasi-transverse* modes. BLS frequency shifts are measured as the material is rotated from one on-axis direction to the next, thus rotating the direction of the phonon wave-vector in the material. With enough such measurements, one can numerically fit the sound velocities to the Christoffel equations and obtain all the independent stiffness tensor elements (and, in some cases, the refractive index). Measured values are fit, usually via a linear least square model, to solve for the individual coefficients. Some common stiffness tensor symmetries that have been used to describe biological samples are shown in Supplementary Figure 5.

An analytic form of the Christoffel equations for a *cubic symmetry* can be obtained. When the wavevector, k , lies in the (001) plane at an angle ϕ relative to the x-axis these are given by:

$$V_L = \sqrt{\frac{\frac{1}{2}\{(c_{11} + c_{44}) \pm [(c_{11} - c_{44})^2 - 4A(c_{11} + c_{12})\cos^2\phi \sin^2\phi]^{1/2}\}}{\rho}}$$

$$V_{T_2} = \sqrt{\frac{\frac{1}{2}\{(c_{11} + c_{44}) \pm [(c_{11} - c_{44})^2 - 4A(c_{11} + c_{12})\cos^2\phi \sin^2\phi]^{1/2}\}}{\rho}}$$

$$V_{T_1} = \sqrt{\frac{c_{44}}{\rho}} \quad \text{Eqn. S19}$$

Isotropic $\begin{pmatrix} c_{11} & c_{12} & c_{12} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{12} & 0 & 0 & 0 \\ c_{12} & c_{12} & c_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2}(c_{11}-c_{12}) & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2}(c_{11}-c_{12}) & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2}(c_{11}-c_{12}) \end{pmatrix}$	Cubic $\begin{pmatrix} c_{11} & c_{12} & c_{12} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{12} & 0 & 0 & 0 \\ c_{12} & c_{12} & c_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{44} \end{pmatrix}$
Hexagonal $\begin{pmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{13} & 0 & 0 & 0 \\ c_{13} & c_{13} & c_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2}(c_{11}-c_{12}) \end{pmatrix}$	

Supplementary Figure 5: Stiffness Tensor Symmetries. Three of the most common stiffness tensor symmetries used to describe biological systems measured with BLS. Note: In the isotropic, cubic and hexagonal cases there are only 2, 3 and 4 independent components respectively.

When on the other hand the wavevector, k , is in the (011) plane at an angle ϕ relative to the x-axis these will be given by:

$$V_L = \sqrt{\frac{\frac{1}{4} \{ (c_{11} + c_{12} + 4c_{44}) + A_1 \cos^2 \phi \pm [(c_{11} + c_{12})^2 - A_1(6c_{11} + 14c_{12} + 8c_{44}) \cos^2 \phi + A_1(9c_{11} + 15c_{12} + 6c_{44}) \cos^4 \phi]^{1/2}}{\rho}}$$

$$V_{T_1} = \sqrt{\frac{\frac{1}{4} \{ (c_{11} + c_{12} + 4c_{44}) + A_1 \cos^2 \phi \pm [(c_{11} + c_{12})^2 - A_1(6c_{11} + 14c_{12} + 8c_{44}) \cos^2 \phi + A_1(9c_{11} + 15c_{12} + 6c_{44}) \cos^4 \phi]^{1/2}}{\rho}}$$

$$V_{T_2} = \sqrt{\frac{\frac{1}{2}[(c_{11}-c_{12})-A_2 \cos^2 \phi]}{\rho}} \quad \text{Eqn. S20}$$

where $A_1 = c_{11} - c_{12} - 2c_{44}$ and $A_2 = c_{11} - c_{12} - 2c_{44}$. In each case one fits angle (ϕ) resolved measurements to extract the different unknown stiffness tensor coefficients. The bulk, Young's and shear elastic moduli can be calculated from:

$$K = \frac{(c_{11}+2c_{12})}{3} \quad \text{Eqn. S21a}$$

$$E = \frac{(c_{11}-c_{12})(c_{11}+2c_{12})}{(c_{11}+c_{12})} \quad \text{Eqn. S21b}$$

$$G = c_{44} = \frac{(c_{11}-c_{12})}{2} \quad \text{Eqn. S21c}$$

The analytical form of the Christoffel equations for a system with *hexagonal symmetry* are given by:

$$V_L = \frac{c_{11}\sin^2\phi + c_{33}\cos^2\phi + c_{44} \pm \sqrt{[(c_{11}-c_{44})\sin^2\phi + (c_{44}-c_{33})\cos^2\phi]^2 + 4(c_{13}+c_{44})^2\sin^2\phi\cos^2\phi}}{2\rho}$$

$$V_{T_1} = \frac{c_{11}\sin^2\phi + c_{33}\cos^2\phi + c_{44} \pm \sqrt{[(c_{11}-c_{44})\sin^2\phi + (c_{44}-c_{33})\cos^2\phi]^2 + 4(c_{13}+c_{44})^2\sin^2\phi\cos^2\phi}}{2\rho}$$

$$V_{T_2} = \sqrt{\frac{c_{66}\sin^2\phi + c_{44}\cos^2\phi}{\rho}} \quad \text{Eqn. S22}$$

Here the bulk, Young's and shear elastic moduli will be given by:

$$K = \frac{-2c_{13}^2 + (c_{11}+c_{12})c_{33}}{c_{11}+c_{12}-4c_{13}+2c_{33}} \quad \text{Eqn. S23a}$$

$$E_{\parallel} = c_{33} - \frac{2c_{13}^2}{c_{11}+c_{12}} \quad E_{\perp} = \frac{(c_{11}-c_{12})[c_{33}(c_{11}+c_{12})-2c_{13}^2]}{c_{11}c_{33}-c_{13}^2} \quad \text{Eqn. S23b}$$

$$G_1 = c_{44} \quad G_2 = c_{66} \quad \text{Eqn. S23c}$$

$$\sigma_{12} = \frac{c_{33}c_{12}-c_{13}^2}{c_{33}c_{11}-c_{13}^2} \quad \sigma_{13} = \frac{c_{13}}{c_{11}+c_{12}} \quad \text{Eqn. S23d}$$

2.8. Registration to accepted/standard values

Different BLS instruments operate at different wavelengths and there are various extrinsic factors that can affect the BLS frequency shift (ν_B) and linewidth (Γ_B). It has previously been proposed that rather than simply reporting ν_B and Γ_B it may be desirable to report $\underline{\nu}_B = \nu_B/\nu_B^w - 1$ and $\underline{\Gamma}_B = \Gamma_B/\Gamma_B^w - 1$, where ν_B^w and Γ_B^w are the BLS frequency shift and linewidth of water measured using the same instrument, at the same temperatures and experimental conditions. The intention thereof was to allow for easy comparison of measurements from different instruments with different probing wavelengths, and to some degree also account for different artifacts that would perturb measurements of the probed sample in a similar way as water. These however do not account for constant systematic offsets or spectral broadening resulting from spectrometer design and measurement geometry. We thereby propose the following correction.

For ν_B a systematic offset can be accounted for by comparing the measured frequency shift for *pure* water (ideally from an injection grade ampoule, else double or triple distilled) $\nu_B^{(w)}$ measured with the same scattering geometry (with effective scattering wavevector $q^{(w)}$, Eqn. S5) as the sample, with an accepted value for water $\nu_B^{(w,0)}$ at the same temperature (T). We propose the following correction for the BLS frequency shift $\nu_B^{(a)}$ measured for sample “a” :

$$\nu_B'^{(a)}(T) = \nu_B^{(a)}(T) - \nu_B^{(w)}(T) + (q^{(w)}/q^{(w,0)})\nu_B^{(w,0)}(T) \quad \text{Eqn. S24}$$

Where $\nu_B^{(w,0)}$ is an accepted standard value of water, and $q^{(w,0)}$ the scattering wavevector of the standard value measurement. From this it follows that the corrected hypersonic speed $V'^{(a)}(T)$ will be given by:

$$V'^{(a)}(T) = (2\pi/q^{(a)}) \left[\nu_B^{(a)}(T) - \nu_B^{(w)}(T) + (q^{(w)}/q^{(w,0)})\nu_B^{(w,0)}(T) \right] \quad \text{Eqn. S25}$$

and the change in the hypersonic speed relative to the corrected value for water can compactly be written as:

$$\frac{V'^{(a)}(T)}{V'^{(w)}(T)} = \frac{q^{(w,0)}}{q^{(a)}} \left[\frac{\nu_B^{(a)}(T) - \nu_B^{(w)}(T)}{\nu_B^{(w,0)}(T)} \right] + \frac{q^{(w)}}{q^{(a)}} \quad \text{Eqn. S26}$$

An analogous correction can be applied to the linewidth:

$$\Gamma_B'^{(a)}(T) = \Gamma_B^{(a)}(T) - \Gamma_B^{(w)}(T) + (q^{(w)}/q^{(w,0)})^2 \Gamma_B^{(w,0)}(T) \quad \text{Eqn. S27}$$

Where $\Gamma_B^{(w,0)}$ is the accepted standard value of water. It follows that the corrected kinematic viscosity (μ_L) will be given by:

$$\mu_L'^{(a)}(T) = (2\pi/q^{(a)2}) \left[\Gamma_B^{(a)}(T) - \Gamma_B^{(w)}(T) + (q^{(w)}/q^{(w,0)})^2 \Gamma_B^{(w,0)}(T) \right] \quad \text{Eqn. S28}$$

and the relative difference of the corrected kinematic viscosity to that of water will be:

$$\frac{\mu_L'^{(a)}(T)}{\mu_L'^{(w)}(T)} = \left(\frac{q^{(w,0)}}{q^{(a)}} \right)^2 \left[\frac{\Gamma_B^{(a)}(T) - \Gamma_B^{(w)}(T)}{\Gamma_B^{(w,0)}(T)} \right] + \left(\frac{q^{(w)}}{q^{(a)}} \right)^2 \quad \text{Eqn. S29}$$

An interpolated table of *standard values* for $\nu_B^{(w,0)}(T)$, $\Gamma_B^{(w,0)}(T)$ and $q^{(w,0)}$ that may be used, are provided in the *Supplementary Table*. These were obtained from deconvolved TFP measurements on injection grade water ampoules in laboratories in Vienna and Hannover. They were performed in $\leq 1^\circ\text{C}$ temperature steps over the range 20-40°C, and throughout showed agreement of better than 0.1% with each other for both the frequency shift and linewidth. The acoustic speed derived from these values agreed with the accepted acoustic

velocity in pure water to within 0.2% over the temperature range 25-35°C for which the latter was available.

2.9. Determination of the refractive index and mass density

As can be seen, calculation of viscoelastic parameters typically requires knowledge of extrinsic parameters (ρ and n), which may not always be accessible. In homogeneous materials, n can be measured using an Abbe refractometer, whereas in thin heterogeneous samples it can be spatially mapped using phase-resolved methods^{53,54}. ρ can be estimated from n via the Lorentz-Lorentz relation and knowledge of the proportionality coefficient(s)⁵⁵ (Supplementary Text §2.10). This however may not be trivial in heterogeneous samples, where these vary spatially. Alternatively ρ may be obtained using the linear gradient method⁵⁶ and for homogeneous samples using a densimeter⁵⁷ or measurement of sample mass and volume. If ρ and n are not accessible, then an approximation may be made from literature values. For samples with a spatially uniform chemical composition and material phase, a near constant ratio of ρ/n^2 may in certain cases be assumed⁵⁸, however this can break down when samples have gradients in composition (e.g. interfaces between hydrophilic-hydrophobic regions).

It may be possible to estimate the mass density from the refractive index (or vice versa) via the *Clausius–Mossotti equation* (or equivalently the *Lorentz-Lorentz equation*). This relates the (macroscopic) optical properties of a material to its microscopic physical properties (molecular polarizability and molecular density), and has been shown to be applicable in diverse solid materials. It predicts a relation between the refractive index (n) and the mass density (ρ):

$$\frac{(n^2-1)}{(n^2+2)} = \left(\frac{4\pi}{3}\right) N\alpha = \left(\frac{4\pi}{3}\right) \left(\frac{\rho}{M}\right) \alpha \quad \text{Eqn. S30}$$

where N is the number density of molecules, M is their molecular weight, and α is the electronic polarizability. When the electronic polarizability or the number density of molecules is small (as $N\alpha \rightarrow 0$), this reduces to: $n^2 - 1 \approx 4\pi M^{-1}\alpha\rho$ and further to $n - 1 \approx 2\pi M^{-1}\alpha\rho$. These approximations have been found to be valid for gases under normal pressure. As can be seen none of these relations directly imply that n^2/ρ is constant (yet alone material independent) as would be desired for the BLS frequency shift to be related directly to the elastic modulus (Eqn. S10). As such independent measurements of both the refractive index n and mass density ρ are in general required to obtain accurate prediction of viscoelastic moduli from BLS data.

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